

The Operational Question (e.g., "Why did efficiency drop last week?").

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## CSV FILE
### mock_ops.csv data
- timestamp
- date
- Zone
- Shift
- Pressure_PSI
- Temperature_C
- Flow_Rate_LPM

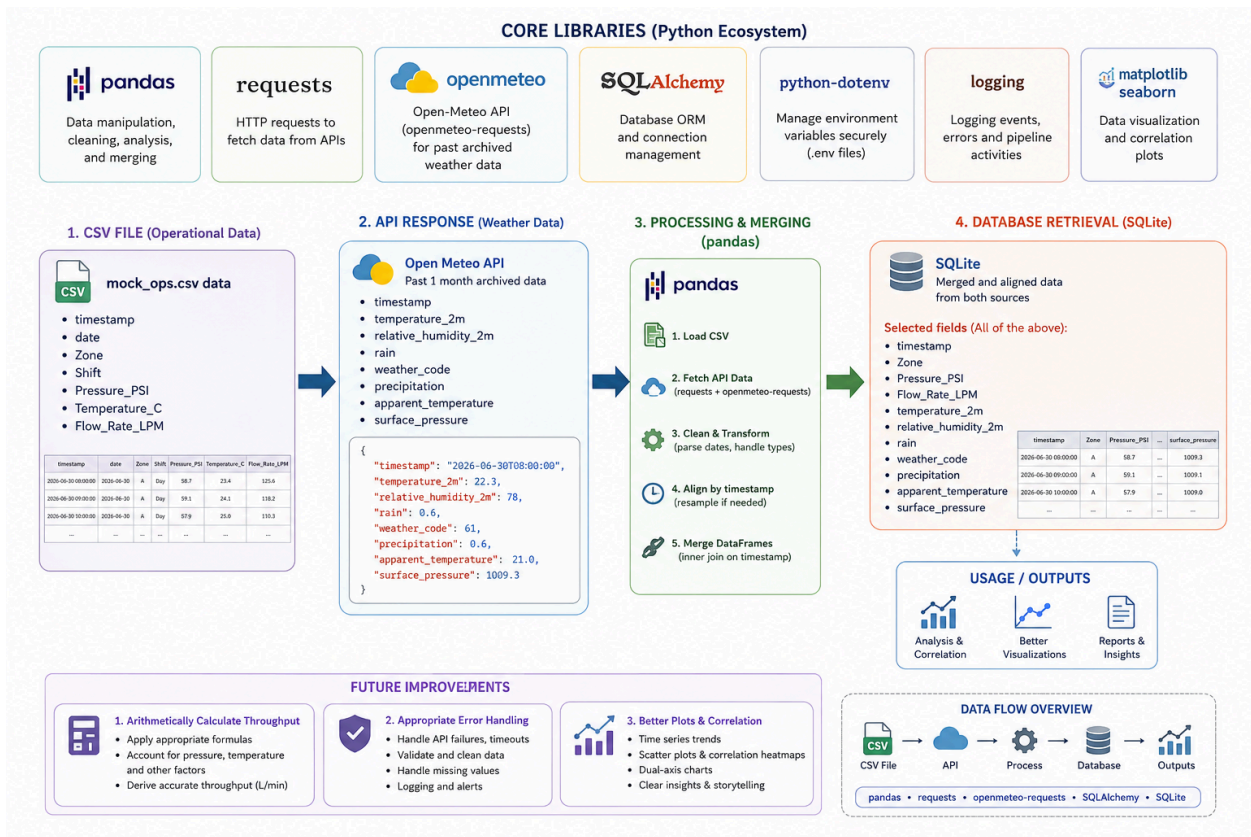
## API RESPONSE
### Open Meteo API - past 1 month archived data
- timestamp
- temperature_2m
- relative_humidity_2m
- rain
- weather_code
- precipitation
- apparent_temperature
- surface_pressure

## DATABASE RETRIEVAL
### Merged and aligned data from both mock_ops.csv and API response
#### All of the above fields, selected:
- timestamp
- Zone
- Pressure_PSI
- Flow_Rate_LPM
- temperature_2m
- relative_humidity_2m
- rain
- weather_code
- precipitation
- apparent_temperature
- surface_pressure

My future improvements:
-- arithmetically calculate throughput with appropriate formulas
-- appropriate error handling
-- better plots, translating the correlation well
```

The Operational Question (e.g., "Why did efficiency drop last week?").

2: The Data Integration (Show a diagram of how you combined API + DB + Internal Data).



3: The Insight & Recommendation (What did the merged data reveal?).

Temperature vs Flow Rate Correlation: 0.01

Zone Pressure_PSI Flow_Rate_LPM \

timestamp

2026-06-25 02:00:00	Zone_East	160.157953	673.194158
2026-06-26 21:00:00	Zone_South	184.513951	716.486706
2026-06-30 08:00:00	Zone_East	205.383753	716.968183
2026-06-27 02:00:00	Zone_Central	201.468924	723.548105
2026-07-01 11:00:00	Zone_Central	201.464269	743.424002

temperature_2m relative_humidity_2m rain weather_code \

timestamp

2026-06-25 02:00:00	19.000000	87.074471	0.0	1.0
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2026-06-26 21:00:00	25.600000	67.763252	0.0	0.0
2026-06-30 08:00:00	20.750000	78.373680	0.0	3.0
2026-06-27 02:00:00	14.600000	78.747032	0.0	0.0
2026-07-01 11:00:00	18.299999	67.588943	0.1	51.0

	precipitation	apparent_temperature	surface_pressure
timestamp			
2026-06-25 02:00:00	0.0	19.571442	931.741638
2026-06-26 21:00:00	0.0	25.459936	1014.429260
2026-06-30 08:00:00	0.0	21.596422	933.054871
2026-06-27 02:00:00	0.0	13.795815	838.362915
2026-07-01 11:00:00	0.1	18.233372	841.232178
Zone	str		

